CMPS/IT 112 – Introduction to Computing and Information Technology (FYDT,FYOC, EP1)

Fall 2019

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Course Number and Title:

CMPS/IT 112 – Introduction to Computing and Information Technology

Course Level:

This course is intended to be taken during the student's first semester of study.

Prerequisites:

None.

Catalog Description:

CMPS/IT 112 – Introduction to Computing and Information Technology 3.0 credits

This course presents foundational concepts and surveys the past, present and future of computing and information technology with an emphasis on the seminal ideas, concepts, inventions and innovations that fuel the Digital Revolution. The sub-disciplines of computing are identified and discussed relative to each other. This course is designed to foster knowledge and abilities needed for gathering, evaluating and disseminating information with an emphasis on digital technology and oral communication.

Student Learning Outcomes:

Upon successful completion of this course students will be able to:

Foundational Concepts:

1. Identify and explain the significance of milestone events in the history of digital technology.
2. List and describe each of the functional components of a computer system.
3. Describe how digital systems represent, store and transmit data and information.
4. Describe how data becomes information and how information is used in decision making.
5. Discuss and evaluate the pros and cons of technological invention and innovation, in particular in regard to the impact it has on people's lives.
6. Understand the significance of and evaluate emerging technological developments.
7. Explain the emphasis of Computer Science, Information Technology and Software Engineering and the interrelationship among the three.

First-Year Digital Technology (FYDT):

1. Conduct effective search strategies to gather information suitable to the topic, audience, purpose, context and speaker.
2. Evaluate sources for credibility.
3. Use digital technology to analyze and process data and information.
4. Employ digital technology to deliver results in appropriate forms.

First-Year Oral Communication (FYOC):

1. Structure and organize information according to purpose, audience, and situation.
2. Develop and share ideas in both formal and informal situations using verbal and non-verbal communication.
3. Create and maintain a relationship between the speaker and audience.
4. Engage in effective listening and self-reflection.

Topics:

1. History of computing, computers and digital technology.
2. Hardware and software components of computer systems.
3. Data representation.
4. Communication and networking.
5. Databases and information systems.

6. Privacy and security issues in the digital world.
7. Computer Science, Information Technology and Software Engineering.

A 94 D 65
B 85 F <65
C 75

Homework, Assignments, and Exams

1. Homework/assignments 30%

   Specific assignments will deal with each of the prime abilities of gathering, evaluating and disseminating information. The context of these assignments will relate to the foundational concepts. The major assignment of the course deals with the development of a Capstone Topic and corresponding Capstone Presentation.

2. Use of Technology 20%

   Students will utilize various software tools in the work of this course, and will be expected to make appropriate use of the features and functionality of each tool, ideally growing their expertise. Included among these tools are word processors, electronic spreadsheets, search engines, and web-based tools.

   Some of the early assignments will be designed as "warm up" assignments meant to evaluate each student's use of particular software tools. The grades on these assignments will be used, in part, to determine the Mid-semester grade, but will not count toward determining the final course grade.

3. Presentations 20%

   Each student will give three individual presentations, and in addition will be expected to actively participate in all classroom discussions which will be of the seminar/workshop format. The third and final presentation, known as the Capstone Presentation, will be the most substantial and will require the student to offer a persuasive argument on some approved technological topic or issue.

4. Quizzes 15%

   Short quizzes will be used to evaluate the student's understanding of assigned readings from each of the sources used.

5. Final examination (date - TBD) 15%

Suggested Textbook and References:

1. An online collection of material on Digital Technology and related issues, available at http://www.cs.scranton.edu/~ep/EP/ep.html
2. A Pocket Guide to Public Speaking by O'Hair, Rubenstein & Stewart, 4th edition or newer.
3. The Weinberg Library's "how-to" research information, available online at https://guides.library.scranton.edu/c.php?g=334470

Disability Policy

Request for Accommodations: Reasonable academic accommodations may be provided to students who submit appropriate and current documentation of their disability. Students are encouraged to contact the Center for Teaching and Learning Excellence (CTLE) at disabilityservices@scranton.edu or (570) 941-4038 if they have, or think they may have, a disability and wish to determine eligibility for any accommodations. For more information, please visit www.scranton.edu/disabilities.

Attendance/Behavior Policy

I expect all students to act in a professional manner. This includes regular attendance, participation in class discussions and proper attire. Frequent absences (>2) will result in a final grade penalty. Unauthorized computer/cell phone use in class is unacceptable.

My Reporting Obligations as a Responsible Employee

It is important for you to know that all faculty members are required to report incidents of sexual harassment or sexual misconduct involving students. This means that I cannot keep information about sexual harassment, sexual assault, sexual exploitation, intimate partner violence or stalking confidential if you share that information with me. I will keep the information as private as I can but am required to bring it to the attention of the University's Title IX Coordinator, Elizabeth Garcia, the Deputy Title IX Coordinator, Christine M. Black, or Deputy Title IX Coordinator/Dean of Students, Lauren Rivera, who, in conversation with you, will explain available support, resources, and options. I will not report anything to anybody without first letting you know and discussing choices as to how to proceed. The University's Counseling Center (570-941-7620) is available to you as a confidential resource; counselors (*in the counseling center*) *do not* have an obligation to report to the Title IX Coordinator.

Non-discrimination Statement

The University of Scranton is committed to providing a safe and nondiscriminatory employment and educational environment. The University does not discriminate on the basis of race, color, national origin, sex, disability, religion, age, veteran status, gender identity or expression, sexual orientation, or other status protected by law. Sexual harassment, including sexual violence, is a form of sex discrimination prohibited by Title IX of the Education Amendments of 1972. The University does not discriminate on the basis of sex in its educational, extracurricular, athletic, or other programs or in the context of employment.

Students who believe they have been subject to sexual harassment, sexual misconduct or sex or gender discrimination should contact Elizabeth Garcia, Title IX Coordinator, at (570) 941-6645 or elizabeth.garcia2@scranton.edu, Christine M. Black, the Deputy Title IX Coordinator, at (570) 941-6645 or christine.black@scranton.edu or Ms. Lauren Rivera, AVP for Student Formation & Campus Life and Dean of Students, at (570) 941-7680 or lauren.rivera@scranton.edu. The United States Department of Education's Office for Civil Rights (OCR) enforces Title IX. Information regarding OCR may be found at www.ed.gov/about/offices/list/ocr/index.html

The University of Scranton Sexual Harassment and Sexual Misconduct Policy can be found online at www.scranton.edu/diversity. All reporting options and resources are available at www.scranton.edu/CARE.